

CST 243-3 Rapid Application Development

Lesson 04: Web Technologies for Rapid Development

Activity 09: Samadhi Bookstore Web Application

Under the *Resources* folder, you can find the main template of the Samadhi web application. Download this template and implement the following tasks. You should follow OOP concepts when implementing this application.

1. Develop a registration form (sign_up.jsp) to collect necessary information from users, such as their first name, last name, email address, and desired username and password.
2. Create a "registration" servlet page to handle the submission of the registration form.
3. Upon successful registration, store the user's information in a database.
4. Redirect the user to the "sign-up.jsp". If the registration is successful, display the message "You have successfully registered with Samadhi Bookstore." Otherwise, show an error message: "An error occurred. Please register again."
5. Create a login form (index.jsp) that prompts users to enter their username and password.
6. Develop the "login" servlet page to process the submitted login form.
7. After a successful login, store the user in a session.
8. Use the "index.jsp" page available under the folder named "user" to serve as the home page for users after a successful login.
9. The user's home page should display a welcome message with their first name and last name, along with a Logout option.
10. If a user attempts to access the user's home directly without logging in, redirect them to the "index.jsp" page where the login form is.
11. Implement a logout function for logged-in users to successfully log out from their user accounts. This logout should redirect the user to the login form page again.

Additional Notes:

- Use following class diagrams and MySQL commands when implementing this Samadhi Bookstore application.

User
- id: int - firstName: string - lastName: string - username: string - password: string - role: string
+User(username: string, password: string) +User(firstName: string, lastName: string, username: string, password: string, role: string) +setId(id: int) : void +getId() : int +getFirstName() : string +getLastName() : string +getRole() : string +register(con: Connection) : boolean +authenticate(con: Connection) : boolean +update(con: Connection) : boolean

Figure 1: User Class

```
CREATE TABLE user (
  id INT PRIMARY KEY AUTO_INCREMENT,
  firstName VARCHAR(255) NOT NULL,
  lastName VARCHAR(255) NOT NULL,
  username VARCHAR(255) NOT NULL,
  password TEXT NOT NULL,
  role VARCHAR(255) NOT NULL COMMENT 'Available roles: admin, customer,
salesman'
);
```